VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE

Matunga, Mumbai-400 019

Autonomous Institute affiliated to University of Mumbai

EXAMINATION	ESE May - June Examination 2024	DATE OF EXAM	24/05/2024
SEMESTER & PROGRAM	Sem-II, First Year MCA	TIME	01.15 pm – 04.15 pm
TIME ALLOWED	**3 HR**	MARKS	**100**
COURSE NAME – (CODE)	Cloud Computing – (R5MC5113S)		

Instructions

1. All questions carry equal marks.
2. Figures to the right indicate full marks.
3. Assume suitable data if required.

Q.1	a.	Differentiate between Vertical scale up and Horizontal Scale out. (At least valid 5 points).	(5)	CO1
	b.	What is the need of Cloud SLA? Explain in detail.	(3)	CO1
	c.	Which are the various aplications where PaaS may be useful?	(5)	
	d.	Consider the peak computing demand for an organization is 200 units. The demand as a function of time can be expressed as: $D(t) = 15\sin(t) \quad 0 <= t < \pi/2$ $\quad\quad 55\sin(t) \quad \pi/2 <= t < \pi$ The resource provisioned by the cloud to satisfy the current demand at time t is given as: $R(t)= D(t) + \delta(d(D(t))/dt)$ Where, δ is the delay in provisioning the extra computing resource on demand. The cost to provision unit cloud resource for unit time is 0.7 units. Calculate the penalty. (Assume the delay in provisioning is $\pi/6$ time units and the minimum demand is 0.)	(5)	CO3
	e.	In a cloud service uptime is 100 minutes and downtime is 20 minutes. What is the availability of service?	(2)	CO2
Q.2	a.	What are Service Level Objectives (SLOs)? Discuss web service SLAs in detail.	(5)	CO3
	b.	What is map-reduce? Apply to the following word-count problem, with the input: F1: This is an Apple. F2: Apple is red in color.	(6)	CO4
	c.	In a map reduce framework consider the HDFS block size is 256MB.	(3)	CO2

		We have 3 files of size 32Kb, 312MB, and 650MB. How many blocks will be created by Hadoop framework?		
	d.	Apply map reduce to the following word-count problem, with the input: F1: Walchand, VJTI, IITBHU F2: PNVK, PNVK, IITBHU F3: Walchand, PNVK, IITBHU	(6)	CO4
Q.3	a.	Consider the peak computing demand for an organization is 400 units. The demand as a function of time can be expressed as $D(t)= 60(1+e^{t})$ Baseline (owned) unit cost is 220 and cloud unit cost is 300. In this situation is cloud cheaper than owning for a period of 300 time units?	(5)	CO3
	b.	Illustrate web service model and Universal Description Discovery and Integration (UDDI) with diagram.	(5)	CO1
	c.	What is the motivation for ~~IoT~~ Sensor cloud? Illustrate ~~IoT~~ Sensor cloud in detail.	(5)	CO3
	d.	Write the pseudo-codes (for map and reduce functions) for calculating the average of a set of integers in MapReduce. Suppose A= (25, 35, 45, 55, 65) is a set of integers. Take 3 mapper nodes and 1 reducer node. Show the map and reduce outputs.	(5)	CO3
Q.4	a.	A third party application runs in the cloud for 11 hours/day. At the end of one month [31 days], it was found that the cloud service suffered 6 outages of durations: 30 minutes, 1 hour 15 minutes, 45 minutes, 2 hours, 1 hour, and H hours, each on different days over the service period. Suppose a cloud guarantees service availability for 97% of time. What will be the possible value of H that SLA negotiation gets honored in terms of service availability?	(4)	CO2
	b.	What are different parallel database architectures? Describe in brief.	(4)	CO1
	c.	Describe key components of Openstack Cloud Architecture.	(6)	CO1
	d.	Write pseudo codes for computing total and average Salary of organization XYZ and grop by Gender (male or female) using MapReduce. The input is as follows: Name, Gender, Salary Jay, M, 50000 Nayana, F, 55000	(6)	CO3
Q.5	a.	A company Y needs to support a spike in demand when it becomes	(6)	CO2

popular followed potentially by a reduction once some of the visitors turn away. The company has two options to satisfy the requirements which are given in the following table:

Expenditures	In House Server	Cloud Server
Purchase Cost	700000	No Cost
Number of CPU cores	11	7
cost / hour(over 5 year span)	No Cost	41
efficiency	45%(underutilized)	85.00%
power and cooling (cost / hour)	21	-
management cost (cost / hour)	5	1

1. Calculate the price of a core-hour on in –house server and cloud server.
2. Find the cost / effective hour for both the option.
3. Calculate the ratio of the total cost / effective hour for in house to cloud deployment.
4. If the efficiency of in house server is increased to 75% which deployment will have now better total cost / effiective hour?
5. If the efficiency of in house server is increased to 90% which deployment will have now better total cost / effective hour?
6. If the efficiency of in house server is increased to 95% which deployment will have now better total cost / effective hour?

	b.	What is the motivation behind IoT cloud computing? Explain IoT cloud computing in detail.	(5)	CO4
	c.	Suppose a cloud guarantees a service availability for 97% of time. Let a third party application runs in the cloud for 9 hours/day. At the end of one month, it was found that total outage is 6.25 hours. Find out whether the proider has violated the initial availability guarantee.	(4)	CO2
	d.	Consider a scenario where a company Y wants to use a cloud service from a provider P. The service level agreement (SLA) guarantees negotiated between the two parties prior to initiating the business are as follows: Avaialbility guarantee 99.5% time over the service period. Service period: 31 days	(5)	CO3

		Max service hours per day: 8 hours Cost: Rs. 80/- per day Service credits are awarded to customers if availability guarantees are not satisfied. Monthly connectivity uptime service level are given as: Monthly uptime connectivity %age Service credit <99.5% 20% <99% 30% However, in reality it was found that over the service period, the cloud service suffered 8 outages of the duration: 5 hrs, 20min, 2 hr 20 min, 20min, 3 hrs 25 min, 35min, 2 hr, 2 hr 20 min each on different days, due to which normal service guarantees were violated. If SLA negotiation are honoured, compute the effective cost payable towards buying the cloud service.		
		All the best		

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EXAMINATION	**ESE May-June 2024**	DATE OF EXAM	27/05/2024
SEMESTER & PROGRAM	**Sem - II, First Year MCA**	TIME	1:15 pm to 4:15 pm
TIME ALLOWED	**3 HRS.**	MARKS	**100**
COURSE NAME – (CODE)	**DESIGN AND ANALYSIS OF ALGORITHM – (R5MC5022T)**		

Instructions
- All questions carry equal marks.
- Figures to the right indicate full marks.
- Assume suitable data if required.

Q.1	a.	A person wants to visit some places. He starts from a vertex and then wants to visit every vertex till it finishes from one vertex, backtracks and then explore other vertex from same vertex. What algorithm he should use?	(2)	CO1
	b.	Which algorithm is used to indicate whether the graph has negative weight cycles or not?	(2)	CO1
	c.	Given two sorted lists of size m and n respectively. What is the number of comparisons needed in the worst case by merge sort algorithms?	(2)	CO2
	d.	An unordered list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum nor minimum is?	(3)	CO4
	e.	How many comparisons are needed by selection sort to sort a 11 item list in the worst case? in the best case? in the average case?	(3)	CO3
	f.	Write pseudo code for solution of Fibonacci series using dynamic programing and explain dynamic programming.	(8)	CO2
Q.2	a.	What is the order of each of the following tasks? Choose from O(1), O(log2 n), O(n), O(nlog2 n), O(n2), O(2n)...etc; each order may appear more than once.) (i) What is the worst case time complexity of quicksort on already sorted data elements with the random element chosen as a pivot? (ii) What is the best case time complexity of selection sort on already sorted data elements? (iii) What is the runtime of Dijkstra's algorithm when the implementation is adjacency matrix. (E = edges; V = vertices) (iv) What is the time complexity of finding the neighbor of all node in a graph when using adjacency matrix? (v)) What is the time complexity of the Breadth First Search using adjacency list is? (vi) What is the space complexity of the Floyd-Warshall algorithm?	(6)	CO2
	b.	Explain the characteristics of an algorithm.	(6)	CO1
	c.	Write the pseudo code of Dijkstra's algorithm.	(8)	CO3
Q.3	a.	Assume that the mergesort algorithm, in the worst case, takes 30 seconds for an input of size 64. Calculate the maximum input size that solves it in 6 minutes?	(4)	CO4
	b.	Explain Big O (Big - Oh) and Ω (omega) asymptotic notation. Big-Oh Notation (O)	(6)	CO3

c. Sort the following elements using Quick sort algorithm. (10) CO4
9, 7, 5, 11, 12, 2, 14, 3, 10, 6

Q.4 a. What is in-place sorting? Give examples of in-place sorting and Not in-place sorting algorithms. (4) CO1

b. Write the time complexity of the following algorithm: (6) CO4

(i)

```
function(int n)
{
    if (n==1)
       return;
    for (int i=1; i<=n; i++)
    {
        for (int j=1; j<=n; j++)
        {
            printf("*");
            break;
        }
    }
}
```

(ii)

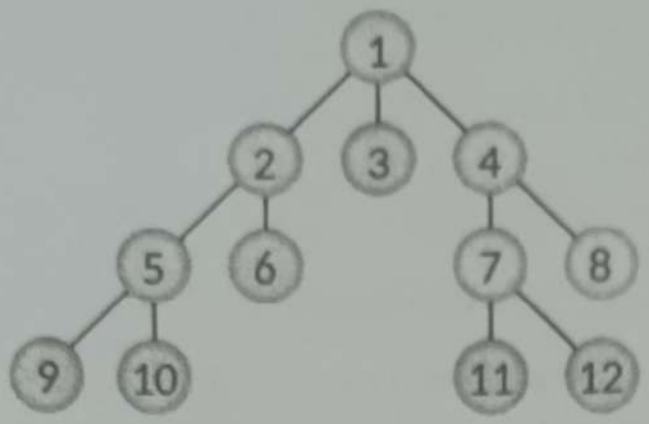
```
#include <stdio.h>
int main() {
    int arr[] = {10, 20, 30, 40, 50};
    int n = sizeof(arr)/sizeof(arr[0]);
    int max = arr[0];
    for (int i = 1; i < n; i++) {
        if (arr[i] > max) {
            max = arr[i];
        }
    }
    printf("Maximum element in the array = %d\n", max);
    return 0;
}
```

(iii)

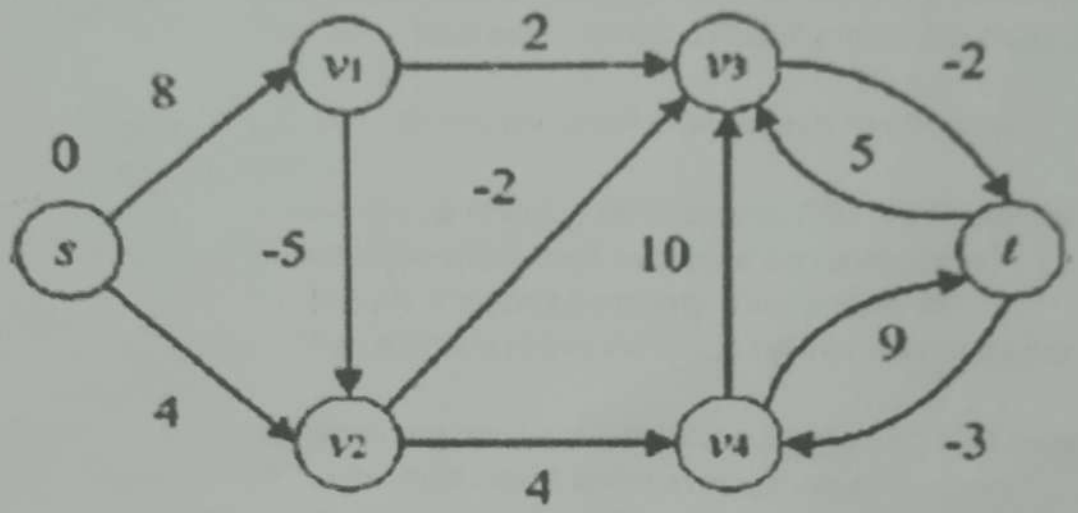
```
int a = 0, i = N;
while (i > 0) {
    a += i;
    i /= 2; }
```

c. Perform BFS traversal on the below graph starting from node ~~A~~ 1. If multiple node choices may be available for next travel, choose the next node in alphabetical order. (10) CO3

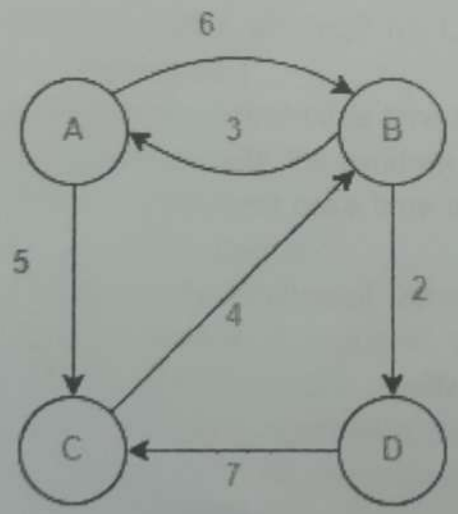
ascending



Q5 a. Find the shortest path from following graph using Bellman ford algorithm. (10) CO2
starting from node S



b. Find the all source shortest path from following graph using Floyd-Warshall algorithm. (10) CO2



VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE (VJTI)
[Central Technological Institute, Maharashtra State]
Matunga, Mumbai-400 019

EXAMINATION	End Semester Examination May 2024	DATE OF EXAM	Wednesday 29.05.2024
SEMESTER & PROGRAM	Semester 2 & MCA	TIME	01.15 PM to 04.15 PM
TIME ALLOWED	3 Hrs	MARKS	100
COURSE (Course Code)	R5MC5123T Software Design and Pattern		

Instructions: All questions are Compulsory

Q. No	Sub-ques.	Question	Marks	CO
Q.1	a	Due to major shift of vehicles on renewable energy, number of vehicles working on electricity in city has increased. Lack of charging stations is the problem identified by local government. In order to address this problem government has decided to set up charging stations, you are appointed as in-charge of this activity. Set up a data model in order to locate charging stations in city.	5M	CO1
	b	Why it is said that "Keep level of Abstraction as high as possible"? How it helps in software design	5M	CO1
	c	Which according to you is better architecture? Data flow architecture or data centered architecture?	5M	CO2
	d	What are different types of Coupling? List and explain each of the type	5M	CO1
Q.2	a	A software is to be built to find out checking the assignments submitted by students, this software will include functionality of checking whether the assignments, laboratory, reports and research paper submitted by students are done with the help of artificial intelligence (like Chat GPT). Perform the use case modelling of the software (assume data).	5M	CO2
	b	What Characteristics of Users must be focused in software development process	5M	CO1
	c	Differentiate between Usability vs. Utility with respect to software design	5M	CO3
	d	What aspects Usability can be divided into, explain with example	5M	CO1
Q.3	a	How various stakeholders in software industry Evaluate User Interfaces	5M	CO3
	b	Explain top down design approach in software architecture design	5M	CO1
	c	List and explain Principles Leading to Good Design	5M	CO3
	d	Explain Communication cohesion with reference to architectural design	5M	CO1
Q.4	a	Institute has opted for giving Krantijyoti Savitribai Phule auditorium for student activity, faculty activity and guest lectures on prior booking basis. It will also have priority in booking. Perform use case modelling for auditorium reservation of VJTI.	5M	CO3
	b	Why do you think that Software Development Should Focus On The Needs Of Users	5M	CO1

	c	List and explain Pattern Description Parameters with reference to software design patterns	5M	CO1
	d	List the various design patterns in Gangs of Four design pattern explain any one.	5M	CO3
Q.5	a	Write code and draw class diagram to explain observer pattern (consider suitable example)	5M	CO3
	b	Perform general class based modelling of delegator pattern also give specific example.	5M	CO4
	c	Give context, forces, problem, solution and anti-pattern for adaptor pattern.	5M	CO4
	d	Give real time example of façade pattern, give class diagram and code for same.	5M	CO3

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VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE

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EXAMINATION	End Semester Examination May June 2024	DATE OF EXAM	31.05.2024
SEMESTER & PROGRAM	Sem-I, FYMCA	TIME	01:15 am to 04:15pm
TIME ALLOWED	**3 HRS.**	MARKS	**100**
COURSE NAME – (CODE)	Mathematical and stastistical foundation II (R5MC5023S)		

Instructions
1. All questions carry equal marks.
2. Figures to the right indicate full marks.

Q.1	a.	Explain prime number and draw the table for 1-100. 5 Write the all (25) prime number up to 100 11 Find the average of first 10 Prime number. 31 How many odd number in 1-100 (1 is not prime number) 35	(5) [CO1]
	b.	Explain composite number and draw the table for 1-100. euler Write all 74 composite number up to 100. theorem Find the average of first 10 composite number. a = 3 n = 10 How many even number in 1-100 (1 is not composite number.)	(5) [CO3]
	c.	What is gradient (slope) and write formula. Write equation of straight line write meaning of y, m, x, c calculate the slope and Draw the straightline (graph) of following value given coordinate 1. $(x_1, y_1) = (4, 2)$ and $(x_2, y_2) = (8, 6)$ 2. $(x_1, y_1) = (2, 1)$ and $(x_2, y_2) = (4, 5)$	(5) [CO4]
	d.	What is GCD find the GCD of following number using simple factor method (i) GCD (30, 52) (ii) GCD (10, 16) (iii) GCD (12, 16) (iv) GCD (12, 8) (v) GCD (24, 36)	(5) [CO1]
Q.2	a.	What is Euclidian method for finding GCD of two number find the GCD of given number i) GCD (33, 12) ii) GCD (900, 750)	(5) [CO3]
	b.	What is M.I. multiplicative inverse of the given number (i.)$2 * ? \equiv 1 \bmod 5$ ii) $2 * ? \equiv 1 \bmod 7$ iii) $5 * ? \equiv 1 \bmod 10$	(5) [CO3]
	c.	What is CRT (Chinese Remainder Theorem)? Solve the following equation using CRT and find the value of X. i. $X \equiv 2 \bmod 3$ ii. $X \equiv 3 \bmod 5$ iii. $X \equiv 2 \bmod 7$	(5) [CO5]

	d.	What is prime factorization of given number? Formula is $n = x*y$; $n = x^2 - y^2$; $x^2 = n + y^2$; $x = sqrt[(n + y^2)]$ and put $y = 1,2,3,.....$ Find the prime factor of $M = 187$	(5) [CO1]
Q.3	a.	What is Regular Expression? Give some examples. Write the 4 rules of Regular Expression: Kleen Closure (*), Positive Closure (+), Concatenation (.), Union (U). Draw the finite automata of the given Regular Expressions: 1) Phi 2) e-epsilon 3) a 4) b 5) ab	(5) [CO1]
	b.	What is CFG context free grammar 1. Given CFG and Generate the string aabb of given grammar CFG = (V, T, P, S) CFG = ({S} {a,b}, P, S) Production rule: 1. S = aSb 2. S=ab 2. Given CFG and generate the string 0101010 of given grammar CFG = (V, T, P, S) CFG=({S}{0,1},P,S Production rule: 1. S = 0S0 2. S=1S1 3. S = 0 4. S= 1 5. S=e-epsilon 3. Given CFG and generate the string baab of given grammar CFG =(V, T, P, S) CFG=({S} {a,b}, P, S Production rule: 1. S= aS 2. S=bS 3. S=a 4. S=b	(5) [CO3]
	c.	What is derivation tree or parse tree 1. Generate the string p*q+r and draw the derivation tree. Given production rule is: E=E+E E=E*E E=p/q/r 2. Generate the string abbbb and draw the derivation tree Given production rule is: S= aXY Y=X Y= e-epsilon X=bYb 3. Generate the string aabbaa and draw the derivation tree Given production rule is: S=aAS S=a A=SbA A=SS A=ba	(5) [CO3]
	d.	What at is modular arithmetic and what are the properties of modular arithmetic and identify the valid or invalid congruence (≡) i) $38 \equiv 2 \bmod 12$ ii) $5 \equiv 0 \bmod 5$ iii) $10 \equiv 2 \bmod 6$	(5) [CO3]

Q.4	a.	What is partial differentiation (derivative) of the following function w.r.t. x 1. d/dx (3yx) 2. d/dx (y^3-x) 3. d/dx (cos x) 4. d/dx (x^4) 5. d/dx ($16x^2$)	(5) [CO4]
	b.	Write notation for higher order derivative and find the different higher order of derivatives of the following function $F(x) = -2x^4 + 10x^3 - 5x^2 + x - 11$ Find the i) f ''(x) ii) f '''(x) iii) f "(1) iv) f '''(2)	(5) [CO4]
	c.	What is gradient descent GD (new = old - n*gradient) in optimization and draw diagram (mountain hill). Explain detail process in GD What are types of gradient descent?	(5) [CO4]
	d.	Prove Euler theorem $a^{\phi(n)} = 1 \bmod n$ hold true for 1. a = 3 and n = 10 2. a = 2 and n = 5	(5) [CO5]
Q.5	a.	What is normal distribution in detail? What are the characteristics of normal distribution? Find the probability density function for the normal distribution where mean = μ = 8 and standard deviation (σ) = 4 and variable x = 6, Pi = 3.14 **Use normal distribution** formula	(5) [CO2]
	b.	Distinguish between DFA and NFA (10 points).	(5) [CO2]
	c.	What is SVM support vector machine? What is hyper plane, support vector margin? What is types of SVM support vector machine? How does working of SVM what is application of SVM?	(5) [CO2]
	d.	What is ML machine learning? Explain detail in different types of ML machine learning? Write the applications of machine learning.	(5) [CO5]

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EXAMINATION	**ESE – Even Semester 2024**	DATE OF EXAM	**03 June 2024**
SEMESTER & PROGRAM	**FY MCA**	TIME	01.15 PM to 04.15 PM
TIME ALLOWED	**3 Hrs**	MARKS	**100**
COURSE (Course Code)	**Professional Communication (R5MC5027T)**		

Instructions: All questions are Compulsory

Q. No	Sub-ques.	Question	Marks
Q.1	a)	Discuss in details the concept and meaning of communication along with differences between verbal and nonverbal communication.	15
	b)	In what ways is public speaking likely to make a difference in your life? How is public speaking similar to every conversation? How is public speaking different from everyday conversation?	15
Q.2	a)	Write a note on the desired behavior during group discussion?	10
	b)	What are the major barriers of communication? Discuss in detail.	10
Q.3	a)	Elaborate the concept of technical writing and style and organization in technical writing.	15
	b)	Discuss in detail the characteristics of an effective presentation.	15
Q.4	a)	Write a note on components of resume writing.	10
	b)	Write a note on preparation for an interview.	10

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EXAMINATION	End Semester Examination May 2024	DATE OF EXAM	Wednesday 05.06.2024
SEMESTER & PROGRAM	Semester 2 & MCA	TIME	01.15 PM to 04.15 PM
TIME ALLOWED	3 Hrs	MARKS	100
COURSE (Course Code)	R5MC5021T Data Mining		

Instructions: All questions are Compulsory

Q. No	Sub-ques.	Question	Marks	CO
Q.1	a	What is Decision Tree? Suppose you have given the following dataset. In this dataset, there are four attributes Age, Competition, Type and Profit, where the ~~Profit~~ Class is a Target attribute or label. On the basis of these attributes, you have to make a Decision Tree. Explain advantages and disadvantages of Decisison Tree.	10M	CO1

SN	Color	Type	Doors	Tires	Class
1	Red	SUV	2	Whitewall	Positive
2	Blue	Minivan	4	Whitewall	Negative
3	Green	Car	4	Whitewall	Negative
4	Red	Minivan	4	Blackwall	Negative
5	Green	Car	2	Blackwall	Positive
6	Green	SUV	4	Blackwall	Negative
7	Blue	SUV	2	Blackwall	Negative
8	Blue	Car	2	Whitewall	Positive
9	Red	SUV	2	Blackwall	Negative
10	Blue	Car	4	Blackwall	Negative
11	Green	SUV	4	Whitewall	Positive
12	Red	Car	2	Blackwall	Positive
13	Green	SUV	2	Blackwall	Negative
14	Green	Minivan	4	Whitewall	Negative

	b	What is the purpose of the linear support vector machine? Suppose we are given the negatively labelled data points ((1,1), (1,-1), (2,1), (2,-1)} and the positively labelled data points {(4,0), (5,1), (5,-1), (6,0)} Find the hyperplane for classification of these points.	10M	CO1
Q.2	a	How to Choose the Value of K in the K-NN Algorithm? Why do we need a KNN algorithm? Apply it to the following new test data instance to classify it properly.	5M	CO2

S. N	x1	x2	Output
1	7	7	0

2	7	4	0			
3	3	4	1			
4	1	4	1			
5	3	7	?			

	b	What are the assumptions of Naive Bayes? Why is it called Naive Bayes?	5M	CO1
	c	What is random forest? What are the various steps Involved in random forest algorithm? Explain in detail.	5M	CO3
	d	What is cluster analysis? Give some real time application of clustering.	5M	CO1
Q.3	a	How to Handle Noisy Data? Explain any one smoothing technique	5M	CO3
	b	List major clustering approaches. Explain partitioning approach with the help of real time example	5M	CO1
	c	Differentiate between two main types of hierarchical clustering methods.	5M	CO3
	d	How do you measure clustering validity via correlation? Explain with an example.	5M	CO1
Q.4	a	List various Data Reduction Strategies, Explain Data cube aggregation.	5M	CO3
	b	What is data warehouse? Give example of a Data Warehouse	5M	CO1
	c	Give conceptual modeling of Data Warehouses, Why ER diagram is used in Data warehouse?	5M	CO1
	d	Explain Data Warehouse Design Process also explain multi tired architecture.	5M	CO3
Q.5	a	What is data mining? Explain knowledge discovery in data process.	5M	CO3
	b	Consider the following table	5M	CO4

Customer ID	Transaction ID	Items Bought
1	1	{a,d,e}
1	2	{a,b,c,e}
2	3	{a,b,d,e}
2	4	{a,c,d,e}
3	5	{b,c,e}
3	6	{b,d,e}
4	7	{c,d}
4	8	{a,b,c}
5	9	{a,d,e}
5	10	{a;b,e}

Taking transaction ID as a market basket find support for each itemset {a}, {c,d}, and {b,d,e}

	c	What are different types of attributes? List and explain properties of attributes.	5M	CO4
	d	How do you measure similarity in clustering? What are types of data in clustering?	5M	CO3